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Abstract This paper is devoted to

Keywords Impulsive reaction-diffusion model, time-space periodicity, maturation delay, traveling waves, spreading speeds, threshold dynamics.

1 Introduction

Reproductive synchrony refers to the temporal clustering of reproductive events among individuals within a restricted time, such as mating, spawning, and births, which is widely recorded in plant and animal populations

2 Model formulation

$$\begin{cases} \dfrac{dT}{dt} = s - \mu T - (1-\eta)\beta_1 T V_I - \beta_2 T I, \\ \dfrac{dI}{dt} = e^{-\mu_1\tau_1}(1-\eta)\beta_1 T(t-\tau_1)V_I(t-\tau_1) + e^{-\mu_1\tau_2}\beta_2 T(t-\tau_2)I(t-\tau_2) - \mu_1 I, \\ \dfrac{dV_I}{dt} = b e^{-\mu_2\tau_3}(1-\epsilon)I(t-\tau_3) - \mu_3 V_I, \\ \dfrac{dV_U}{dt} = b e^{-\mu_2\tau_3}\epsilon I(t-\tau_3) - \mu_3 V_U, \end{cases} \tag{2.1}$$

where s is the rate that the cells reproduce itself and μ is the death rate of the target cell and assume that $\mu_1 \geqslant \mu$.

3 Threshold Dynamics

In this section, we first introduce the basic reproduction ratio R_0 and then study the global dynamics of 2.1. It is easy to see that the scalar linear equation has a unique positive equilibrium $T^* = \frac{s}{\mu}$ and it is globally attractive in $\mathbb{R}$. Linearizing system 2.1 at the virus-free solution

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$(T^*, 0, 0, 0)$, we obtain the following linear system:

$$\begin{cases} \dfrac{dI}{dt} = e^{-\mu_1\tau_1}(1-\eta)\beta_1 T^* e^{-\mu_1\tau_2} V_I(t-\tau_1) + \beta_2 T^* I(t-\tau_2) - \mu_1 I(t), \\ \dfrac{dV_I}{dt} = b e^{-\mu_2\tau_3}(1-\epsilon) I(t-\tau_3) - \mu_3 V_I. \end{cases} \tag{3.1}$$

Let $\tau = \max\{\tau_1, \tau_2, \tau_3\}$, $C = C([-\tau, 0], \mathbb{R}^2)$, and $C^+ = ([-\tau, 0], \mathbb{R}^2_+)$. Then (C, C^+) is an ordered Banach space equipped with the maximum norm and the positive cone C^+. For any continuous function $H : [-\tau, \sigma) \to \mathbb{R}^2$ with $\sigma > 0$, we define $H_t \in C$ by $H_t(\theta) = H(t+\theta)$, $\forall \theta \in [-\tau, 0]$, and any $t \in [0, \sigma)$. Let F, V be two 2×2 matrix defined as follows:

$$F = \begin{bmatrix} \beta_2 T^* e^{-\mu_1\tau_1} - \mu_1 & (1-\eta)\beta_1 T^* e^{-\mu_1\tau_2} \\ 0 & 0 \end{bmatrix}, V = \begin{bmatrix} 0 & 0 \\ -b e^{-\mu_2\tau_3}(1-\epsilon) & \mu_3 \end{bmatrix}, \tag{3.2}$$

Then the linear system 3.1 can be written as

$$\frac{dH(t)}{dt} = FH_t - VH(t). \tag{3.3}$$

Let $X = C([-\tau, 0], \mathbb{R}^4_+)$. Then we have the following result for system 2.1.

Theorem 3.1. *We have the following theorem,*

(i) If $\mathcal{R}_0 < 1$, then the disease-free equilibrium is globally attractive for system 2.1 in X.

(ii) If $\mathcal{R}_0 > 1$, then there is an $m > 0$ such that any solution

$$u(t, \phi) = (T(t, \phi), I(t, \phi), V_I(t, \phi), V_U(t, \phi))$$

of system 2.1 with $\phi \in X$ with $\phi_2(0) > 0$, $\phi_3(0) > 0$ satisfies $\liminf_{t\to\infty}(I(t, \phi), V_I(t, \phi)) > (m, m)$.

Proof. Take the solution map of (3.1) be $Q_t(\phi) = u(t, \phi)$ with $\phi \in C$. Let $Q = Q_1$. By [1, Theorem 6.1, Chapter 3], we have $Q(t)$ is compact for any $t \geqslant \tau$.

In the case where $\mathcal{R}_0 < 1$, let $Q_\gamma(t)$ be the solution map of the following perturbed linear equation on C:

$$\begin{cases} \dfrac{dI}{dt} = e^{-\mu_1\tau_1}(1-\eta)\beta_1 (T^*+\gamma) e^{-\mu_1\tau_2} V_I(t-\tau_1) + \beta_2 (T^*+\gamma) I(t-\tau_2) - \mu_1 I(t), \\ \dfrac{dV_I}{dt} = b e^{-\mu_2\tau_3}(1-\epsilon) I(t-\tau_3) - \mu_3 V_I. \end{cases} \tag{3.4}$$

and $\mathcal{R}_0^\gamma$ is the basic reproduction number for system (3.4). Since $\lim_{\gamma\to 0} \mathcal{R}_0^\gamma < 1$, we can fix a sufficiently small number $\gamma > 0$ such that $\mathcal{R}_0^\gamma < 1$. It is easy to verify that $Q_\gamma(t)$ is also compact and strongly monotone on C for each $t > \tau$. We can see that there is a positive function $v_\gamma(t) = (v_{\gamma,1}(t), v_{\gamma,2}(t))^T$ such that $u_\gamma(t) = e^{\mu_\gamma t} v_\gamma(t)$ is a positive solution of system (3.4) with $\mu_\gamma = \ln \mathcal{R}_0^\gamma$. Clearly, $\lim_{t\to\infty} u_{\gamma,1}(t) = 0$. By the global stability of T^*, there exists a sufficiently large integer $N_1 > 0$ such that $N_1 \geqslant \tau$ and $T(t) < T^* + \gamma$, $\forall t \geqslant N_1 - \tau$. The solution $I(t)$ satisfies this following inequality:

$$\frac{dI}{dt} \leqslant e^{-\mu_1\tau_1}(1-\eta)\beta_1 (T^*+\gamma) e^{-\mu_1\tau_2} V_I(t-\tau_1) + \beta_2 (T^* I + \gamma)(t-\tau_2) - \mu_1 I(t),$$

for $t \geqslant N_1$. Choose a sufficiently large number $k > 0$ such that $I(t) \leqslant ku_\gamma(t)$, $\forall t \in [N_1, N_1+\tau]$. Thus, the comparison theorem for differential equations implies that $I(t) \leqslant ku_\gamma(t)$, $\forall t \geqslant N_1+\tau$. Thus $\lim\limits_{t\to\infty} I(t) = 0$. Taking this result back to the third equation and forth equation of (2.1), we have $\lim\limits_{t\to\infty} V_I(t) = 0$, and $\lim\limits_{t\to\infty} V_U(t) = 0$. Let $w(t) := T(t) - T^*$. In view of (2.1), we have

$$w'(t) = -\mu w - (1-\eta)\beta_1(w + T^*)V_I - \beta_2(w + T^*)I.$$

Since $\lim\limits_{t\to\infty} V_I(t) = 0$ and $\lim\limits_{t\to\infty} I(t) = 0$, it follows that $\lim\limits_{t\to\infty} w(t) = 0$. Thus, $\lim\limits_{t\to\infty} T(t) - T^* = 0$. This proves statement (i).

In the case where $\mathcal{R}_0 > 1$, we will apply the persistence theory for this semiflows. Take

$$X_0 = \{\phi = (\phi_1, \phi_2, \phi_3, \phi_4) \in X : \phi_2(0) > 0,\ \phi_3(0) > 0\}$$

and

$$\partial X_0 = X \setminus X_0 = \{\phi \in X : \phi_2(0) = 0,\ \text{or}\ \phi_3(0) = 0\}.$$

It is easy to see that X_0 is an open set relative to X. Let $\Phi(t) : X \to X(\forall t \geqslant 0)$ be the solution semi-flow generated from system 3.1. Taking any $\phi \in X$, we conclude from the firsts equation of system 2.1 that $\dfrac{dT}{dt} = s - \mu T - (1-\eta)\beta_1 T V_I - \beta_2 T I$. It follows that $T(t,\phi) = \phi_1(0)\exp\left(-\int_0^t \mu + (1-\eta)\beta_1 V_I - \beta_2 I \mathrm{d}\xi\right) + \int_0^t s \exp\left(-\int_m^t \mu + (1-\eta)\beta_1 V_I - \beta_2 I \mathrm{d}\xi\right)\mathrm{d}m > 0$ for all $t > 0$. Similarly, we can get $V_U(t,\phi) = \phi_4(0)e^{-\mu_3 t} + \int_0^t b\epsilon e^{-\mu_3(t-s) - m u_2 \tau_3} I(s-\tau_3)\mathrm{d}s \geqslant 0$, for all $t \geqslant 0$. Since $I'(t) > -\mu_1 I$ and $V_I'(t) > -\mu_3 V_I$ and $\phi_2(0) > 0$, $\phi_3(0) > 0$, it follows that $I(t,\phi) > 0$, $V_I(t,\phi) > 0$ for all $t \geqslant 0$. Thus we have the solution semiflow $\Phi(t) : X_0 \to X_0$. By the uniform and ultimate boundedness of the solution $u(t,\phi)$, it follows that $\Phi(t)$ is point dissipative. By Hale and Verduyn Lunel([1, Theorem 3.6.1]), for each $t > \tau$, $\Phi(t)$ is compact. We prove that $\Phi(t)$ is uniformly persistence with respect to $(X_0, \partial X_0)$. Let $\mathcal{R}_0^m$ be the basic reproduction number of the following perturbed linear equation:

$$\begin{cases} \dfrac{dI}{dt} = e^{-\mu_1\tau_1}(1-\eta)\beta_1(T^* - m)e^{-\mu_1\tau_2}V_I(t-\tau_1) + \beta_2(T^* - m)I(t-\tau_2) - \mu_1 I(t), \\ \dfrac{dV_I}{dt} = be^{-\mu_2\tau_3}(1-\epsilon)I(t-\tau_3) - \mu_3 V_I. \end{cases} \tag{3.5}$$

Since $\lim\limits_{m\to 0} \mathcal{R}_0^m > 1$, we can fix a sufficiently small $m > 0$ such that $\mathcal{R}_0^m > 1$. Let $M_1 = (T^*, 0, 0, 0)$.

Claim $\limsup\limits_{n\to\infty} \|\Phi_t(\phi) - M_1\| \geqslant m$, where $\|\cdot\|$ denotes the Euclidean norm.

Suppose by contradiction, that $\limsup\limits_{n\to\infty} \|\Phi_t(\phi) - M_1\| < m$ for some $\phi \in X_0$. Then there exists an $N_2 > 0$ such that $\|\Phi_t(\phi) - M_1\| < m$ for all $t \geqslant N_2$. Hence,

$$\|\Phi_t(\phi) - \Phi_t(M_1)\| = \|\Phi_t - M_1\| < m,\ t > N_2.$$

It follows that $T(t-\tau) > T^* - m$ for all $t \geqslant N_2 + \tau$. Thus, $u(t,\phi) = (T(t,\phi), I(t,\phi), V_I(t,\phi), V_U(t,\phi))$ satisfies

$$\begin{aligned} \frac{dI}{dt} =& e^{-\mu_1\tau_1}(1-\eta)\beta_1 T(t-\tau_1)V_I(t-\tau_1) + e^{-\mu_1\tau_2}\beta_2 T(t-\tau_2)I(t-\tau_2) - \mu_1 I, \\ \geqslant& e^{-\mu_1\tau_1}(1-\eta)\beta_1(T^* - m)V_I(t-\tau_1) + e^{-\mu_1\tau_2}\beta_2(T^* - m)I(t-\tau_2) - \mu_1 I, \end{aligned}$$

for all $t \geqslant \tau$. Note that $\mathcal{R}_0^m > 1$. As discussed earlier, there is a positive function $v_m > 0$ such that $u_m(t) = e^{\mu_m t} v_m(t)$ is a positive solution of system (3.5) with $\mu_m = \ln \mathcal{R}_0^m$. Sine $\Phi(t)X_0 \subset X_0$, $I(t) > 0$ for all $t \geqslant 0$. We can choose a sufficiently small $k > 0$ such that $I(t) > ku_m(t)$, $\forall t \in [N_2 + \tau, N_2 + 2\tau]$. By the comparison theorem for delay differential equations, it follows that $I(t) \geqslant ku_m(t)$, $\forall t \geqslant N_2 + \tau$. Clearly, $\lim_{t\to\infty} u_m(t) = \infty$. Thus, $\lim_{t\to\infty} I(t) = \infty$, which is a contradiction. This completes the statement (ii). $\square$

4 Numerical simulations

In this section,

5 Discussion

In this paper,

Availability of data and materials

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Acknowledgements

The research supported in part by the NSERC of Canada (RGPIN-2019-05648).

References

[1] J. K. Hale, Theory of Functional Differential Equations, Springer-Verlag, New York, 1977.